

Threats to identification

Slide route: Lecture8v2.tex: external/internal validity, measurement error, missing data, selection, simultaneity, SEs and prediction. Chapter 9.

Session 11 | Diagnose before choosing a remedy

Outcomes: Separate internal from external validity; explain classical attenuation, selection, and simultaneity; match threats to plausible responses.

100-minute route

0–10	Post-midterm retrieval: interpretation versus identification.
10–25	External and internal validity; discuss slide examples.
25–45	OVB, wrong functional form, measurement error.
45–55	Pause and attenuation calculation.
55–75	Missing data, selection, simultaneity.
75–90	Inconsistent SEs, validity checklist, prediction.
90–100	Exit diagnosis and bridge to IV.

Worked example | Measurement error

Use with slides Lecture8v2 slides 12–15: Errors-in-variables and Classical Measurement Error. Work the attenuation calculation on slide 14 or 15, after students have seen why the measured regressor contains noise.

Suppose the true relationship is $Y = \beta_0 + 2X^* + u$, but we observe $X = X^* + v$. Under classical regressor measurement error, v is mean zero and uncorrelated with X^* and u . If $\text{Var}(X^*) = 9$ and $\text{Var}(v) = 3$:

$$\text{plim } \hat{\beta}_{\text{OLS}} = 2 \frac{9}{9+3} = 1.5.$$

Ask Does a larger sample restore the coefficient to 2? No. It concentrates the estimator around the attenuated target. Better measurement or a valid instrument addresses a different problem from increasing precision.

Qualification Attenuation toward zero is a result for this simple classical setup; do not generalize its direction to arbitrary measurement error or every coefficient in multiple regression. Classical noise in the outcome alone need not bias slopes, though it raises uncertainty.

Cases for discussion | Threat, mechanism, response

Selection Wages are observed only among employed people. If employment depends on unobserved wage determinants, the observed sample need not represent the wage-offer relationship for everyone. More complete data or a justified selection design is needed; merely adding a robust SE is insufficient.

Simultaneity Price and quantity are jointly determined by demand and supply. Their observed covariance need not trace a demand curve. A supply shifter excluded from demand is a possible instrument, provided its exclusion and relevance can be defended.

External validity An internally valid tutoring experiment at one university need not identify the same effect for all schools. Ask what differs: students, implementation, context, or treatment intensity.

Dependence Repeated outcomes for one district can be correlated. Heteroskedasticity-robust SEs alone are insufficient; a dependence-aware variance estimator may be needed.

Pair activity | Classify three claims

1. A treatment improves outcomes in one city, so it must do so everywhere.
2. Self-reported education contains substantial recall error.
3. A high- R^2 demand regression uses equilibrium prices.

Answers: External validity; potential errors-in-variables; simultaneity despite fit. For each, ask what extra data or design evidence would make the claim more credible.

Missing data caution Dropping incomplete rows is not automatically harmless. Whether complete-case analysis is valid depends on why data are missing and the estimand. Avoid treating all missingness as one problem with one remedy.

Prediction distinction An endogenous variable can be predictive in a stable environment, yet policy interventions can change its relationship to the outcome. Evaluate predictions out of sample; evaluate causal claims through design and assumptions.

If short / preparation Use one external-validity illustration and three internal threats in depth. Read Chapter 12 and come prepared to propose an instrument and criticize its exclusion restriction.

Source: Lecture8v2.tex; numerical and discussion examples are illustrative.